

PRODUCT OVERVIEW

VDUX360 is an AI-powered decision support application for vacuum distillation units. Domain and machine-learning agents benchmark column, heater and ejector performance, turning historian and DCS data into ranked, quantified recommendations — from heater and pump-around to cut points and valve health. Each deployment is configured and commissioned by Ingenero process engineers to match the unit's feedstock and envelope.



01 Unit Configuration

Shown: column configuration, then model selection against the assets.

VACUUM DISTILLATION COLUMN Load Defaults

NUMBER OF PACKED BEDS: 5 NUMBER OF SIDE DRAWS: 5

Specify Side Draws (Top to Bottom):

Draw-1

Vacuum Diesel

IBP (°C): 340 FBP (°C): 390

Stripper: ☐ Reducible: ☒ Pump Around Reflux: ☒

NO. OF EXCHANGERS IN PUMP AROUND LOOP: 2

Draw-2

LVGO

IBP (°C): 350 FBP (°C): 440

Stripper: ☐ Reducible: ☒ Pump Around Reflux: ☒

NO. OF EXCHANGERS IN PUMP AROUND LOOP: 2

Model Selection:

Live Benchmarking Machine

Calculation Package

Pumparound Optimizer

Asset Reliability

Pinch Analysis

Valve Diagnostics

GenAI Copilot

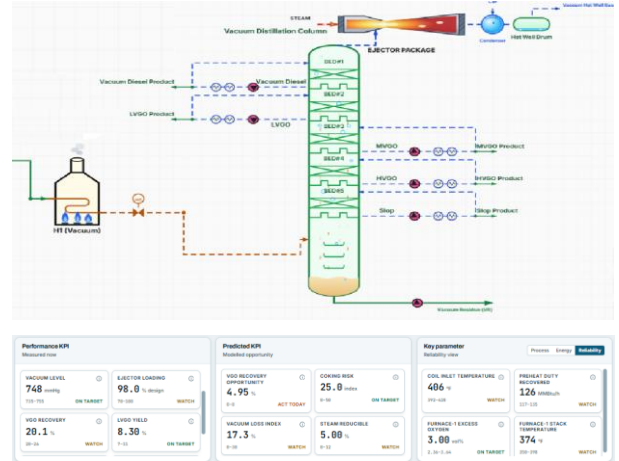
Insights & Suggestions

Asset hierarchy	preheat train, vacuum heater, column, ejector system, pumparounds
Column internals	tray count, side draws and feed tray location
Side draws	product cut, draw-off tray, IBP and FBP per draw
Operating modes	normal, turndown and crude switch; KPIs per asset



02 Plant Data and Model Execution

Shown: the configured unit as a process flow, with live KPIs and trends.



Inputs	crude API, feed rate, COT, draw rates, pumparound/ejector duty
Processing	tag validation, steady state framing and mode segmentation
Outputs	predicted KPIs, ranked actionables with cause, trends, alerts



03 AI-Powered Model Agent Suite

A site enables only the models its assets and available data can support.

Live Benchmarking	Benchmarks column performance against historical optimum.
Vacuum Heater Optimizer	Minimizes excess O ₂ and Stack losses improving fuel consumption.
Pumparound Optimizer	Improves heat recovery and minimizes fuels consumption in heater
Asset Reliability	Outlier detection flags anomalies and failure risk.
Cut Point Optimizer	Improves distillate yield while maintaining product specification
Ejector Performance Monitor	Tracks ejector loading and motive steam
Pinch Analysis	Targets preheat train heat recovery against the pinch.
Valve Diagnostics	Tracks control valve health and ranks tuning gaps.
GenAI Copilot	Answers plant questions inside the operating workflow.
Insights & Suggestions	Quantified actions reported against operating envelope deviations.
Column Performance	Detects flooding, loss of vacuum and increase in ΔP.
Product Quality Soft Sensors	Provides visibility on quality parameters to minimize giveaways.



04 Opportunity and Operator Actions

BUSINESS KPI	CAUSE MESSAGE	ACTUAL	OPTIMUM	SUGGESTION
LOW VGO YIELD	LOW VDU HEATER OUTLET TEMPERATURE	743 °F	748 °F	INCREASE VDU HEATER OUTLET TEMPERATURE TO INCREASE VGO RECOVERY.
	HIGH VDU COLUMN PRESSURE	45 MMHG	40 MMHG	REDUCE VDU COLUMN PRESSURE BY OPTIMIZING EJECTOR OPERATION TO IMPROVE VACUUM AND VGO YIELD
HIGH FUEL GAS CONSUMPTION	HIGH HEATER EXCESS O ₂	3.8 %	2%	REDUCE EXCESS O ₂ TO REDUCE FUEL GAS CONSUMPTION WHILE MAINTAINING SAFE COMBUSTION.
LOW HEATER THERMAL EFFICIENCY	HIGH DRAFT PRESSURE	-0.35 INWC	-0.15 INWC	REDUCE HEATER DRAFT. THIS WILL HELP IN IMPROVING HEATER THERMAL EFFICIENCY

VGO recovery: +0.5–2 wt %

Cut-point optimization

Vacuum heater energy saving: 2–5 %

Excess O₂ and stack losses reduction

Vacuum-system steam saving: 2–5%

Ejector performance optimization

Column throughput: +1–3%

Pump around and column optimization



05 Platform

Historian connectivity connects directly with leading plant historians	Simulation integration third-party simulators integrate without custom engineering	Daily reporting downloadable daily reports with interactive trends	GenAI Copilot answers plant questions in the operating workflow	Build your own dashboard engineers assemble the views each unit needs
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DELIVERY

Configured, validated and commissioned by Ingenero process engineers.

FLEXIBILITY

Any crude slate; selectable KPI set; selectable virtual analyzers.

SCALE

Repeat sites reuse the configuration, with no/minimal rework per plant.

Contact for a free trial
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